

1 A national inventory of well lithology and hydraulics for
2 the conterminous United States, with a vision-language
3 method for reading scanned driller's logs

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5 **Abstract**

6 Sustainable groundwater management requires a three-dimensional picture of the sub-
7 surface, yet across much of the world that information exists only as water-well driller's
8 logs in fragmented, partly-digitized government archives — a barrier most acute where
9 agency-scale digitization is unaffordable. Using the United States as a demonstration,
10 we (i) compile an availability census of the ~ 13.4 million wells in the 48 contermi-
11 nous states, classifying each by how its lithology is published: a first-pass audit finds
12 machine-readable lithology in 8 states (~ 3.2 M wells), but systematic probing raises
13 this to 17 states (~ 7.0 M); 13 states (~ 4 M) publish it only as scanned images, and 18
14 (~ 2.4 M) publish no digital lithology (the wells are registered, the logs are not); (ii) from
15 the machine-readable states we harmonize the ~ 3.4 million wells carrying a log into an
16 open inventory of nearly 24 million depth-resolved intervals, plus structured hydraulics
17 for ~ 2.8 million wells, all from public endpoints, no privileged access; and (iii) intro-
18 duce a reproducible, cost-bounded method that recovers lithology from scanned logs

using vision-language models. Validated against a state agency’s hand-transcriptions, the method reproduces material class at 91.5% accuracy (Cohen’s $\kappa = 0.90$) and the coarse/fine texture that groundwater models consume at 96.0% ($\kappa = 0.92$), at roughly 1,000 tokens per log. We report cost in tokens, a provider-independent unit, and quantify what each state does and does not expose. Because the method transfers to any jurisdiction with scanned well records, it offers a low-cost path to the subsurface data groundwater management needs; data and method are released openly.

Keywords: groundwater; driller’s logs; lithology; vision-language models; subsurface characterization; data harmonization; hydrogeology

Highlights

- Much U.S. well lithology is in scanned driller’s logs, not machine-readable data
- We harmonize 24M lithology intervals from public sources into one open inventory
- Vision-language models recover lithology from scans at 96.0% coarse/fine accuracy
- Cost is about 1,000 tokens per log, using only public, no-privilege access
- The method transfers to any region that holds scanned water-well records

1 Introduction

Continental and coupled groundwater models require the subsurface’s vertical structure — the sequence of geologic materials and their hydraulic properties with depth. In the United States that information is generated as a regulatory byproduct: when a well is drilled, the driller records a log of the materials encountered. Hundreds of millions of such logs exist, but they are held by individual state agencies in heterogeneous, often non-machine-readable forms. Interoperability standards such as GroundWaterML2 exist to bridge these schema differences (Brodaric et al., 2018), but a standard cannot substitute for the data itself: no harmonized national compilation of their lithology exists. Every regional model therefore

re-digitizes its basin’s driller’s logs by hand. Our contributions are: an availability census; a harmonized digital lithology inventory from primary state sources; and a reproducible, cost-bounded method to recover the scanned remainder, with per-state coverage, accuracy, and cost.

This inventory is complementary to the United States Groundwater Well Database (USGWD; Lin et al., 2024), which harmonizes 14.26M well records to a data standard of location, purpose, status, well depth, screen depth/length, and well capacity — but, by design, carries no lithology and no water levels. USGWD is the national well-location layer; this inventory is the subsurface-structure-and-hydraulics layer that USGWD omits. The two compose. Likewise, national hydrogeologic frameworks such as the USGS Secondary Hydrogeologic Regions partition the conterminous US into terrane units at the land surface (Belitz et al., 2019); our depth-resolved lithology populates the third dimension those map-based frameworks leave empty.

1.1 Who needs this, and the cost of not having it

Three independent lines of evidence establish demand. First, the continental-modeling community states the lack explicitly. A flagship review notes that “groundwater models require spatially continuous and consistent datasets of the hydrogeologic properties that control the movement of water through the subsurface,” yet “the identification of hydrogeologic heterogeneity structure and physical properties is difficult because it requires subsurface borehole . . . data that are often scarce” (Condon et al., 2021); de Graaf et al. (2020) state that “no consistent 3D lithologic, hydrogeologic or hydrofacies data set exists” at continental scale; and ParFlow-CONUS is described by its own developers as data-limited (O’Neill et al., 2021). Even USGWD—the harmonized national well-location database—explicitly excludes lithology “due to [its] limited availability” (Lin et al., 2024), naming the precise gap this inventory fills. Second, agencies act on the need: the USGS mined ~458,000 driller’s logs for a single regional flow model (Arihood, 2009) and standardized ~14M state records across

Table 1: Demand by duplicated effort: representative studies that each manually re-compiled water-well driller’s logs because no harmonized national lithology layer existed.

Study	Region	Logs compiled	Purpose
Faunt et al. (2010)	Central Valley, CA	~8,500	3-D sedimentary-texture framework (CVHM)
Arihood (2009)	Lake Michigan Basin	~458,000	glacial ground-water flow model
Vahdat-Aboueshagh et al. (2021)	SW Louisiana	~30,000	56-layer aquifer conductivity framework
Riddle et al. (2025)	MI/IN/OH glacial	~60,500	volumetric hydrogeologic framework
Herzog et al. (2003)	Central Illinois	>2,200	hydrostratigraphic model for MODFLOW
Bayless et al. (2017)	glaciated US (24 states)	~14,000,000	cross-state hydrogeologic grids

the 24 glaciated (northern-tier) states (Bayless et al., 2017), and a federal land manager (USDA Forest Service) grounds its Groundwater Inventory, Monitoring & Assessment guide in hydrogeologic-unit characterization while acknowledging that map-unit lithology “can change laterally” below mappable resolution. Third, and most telling, is duplicated effort: because no harmonized layer exists, every team that needs subsurface lithology re-compiles raw driller’s logs from scratch (Table 1)—thousands to hundreds of thousands of logs per study, each in a bespoke workflow. That recurring, state-by-state re-compilation is the cost of the missing layer, and the demand this inventory answers.

1.2 How continental models represent the subsurface today — and why it is a gap

If primary driller-log lithology is so sparse, how are CONUS-scale subsurface models built? Not from it. Continental hydrologic platforms take one of two routes, neither using harmonized well logs. (1) Conceptual collapse: operational water-balance models—the NOAA National Water Model (WRF-Hydro) and the USGS National Hydrologic Model (PRMS)—resolve only a ~ 2 m soil column and abstract everything beneath into lumped, calibrated reservoirs with no geologic structure. (2) Generalized, map-derived inputs: physically based models parameterize the subsurface from a small, shared set of products derived from geologic maps, not boreholes. The flagship integrated model, ParFlow-CONUS, resolves the entire deep subsurface as one (CONUS1: 5 layers, 102 m; Maxwell et al., 2015) to six (CONUS2: 10 layers, 392 m; Yang et al., 2023) broad units, with conductivity assigned from the global GLHYMPS permeability map (itself derived from the GLiM lithology map, Gleeson et al., 2014; Hartmann and Moosdorf, 2012), soils from SSURGO, and depth-to-bedrock from Shangguan et al. (2017). Its developers state plainly that the model is “limited not by computational expense but by data availability, with a lack of detailed depth-to-bedrock and aquifer thickness estimates at meaningful resolution,” and is better viewed as a “shallow aquifer storage model” (O’Neill et al., 2021). The lithology underneath these permeability products is itself a coarse, surface, polygon map: the GLiM lineage descends from continental compilations such as the 262,111-polygon North American lithological map of Moosdorf et al. (2010), whose authors showed that map *resolution and source alone* shift modeled fluxes by roughly -59% to $+38\%$ — generalization with a measured cost. The same few products recur field-wide (Fan et al., 2013; de Graaf et al., 2017; Pelletier et al., 2016; Hengl et al., 2017); the modelling community has itself begun cataloguing this shared-input homogeneity and its biases through community model portals (Zamrsky et al., 2025).

National geologic products are no better as a three-dimensional lithology source. The seamless USGS State Geologic Map Compilation (Horton et al., 2017) is a surface poly-

gon geodatabase. USGS three-dimensional geologic framework models resolve depth but are regional: an inventory found 38 models covering $\sim 49\%$ of the CONUS at a $\sim 4,170 \text{ km}^2$ median footprint (U.S. Geological Survey, 2023). The one effort to mine bulk driller’s records at scale ($\sim 14\text{M}$ logs; Bayless et al., 2017) is confined to the glaciated 24 states and outputs derived hydrogeologic-property grids, not a lithology layer. At the opposite extreme, research effort yields exquisite per-borehole records — full geophysical and lithologic logs, even NMR-measured hydraulic conductivity — but only at a handful of wells (e.g., Twining et al., 2017; Uhlemann et al., 2019, $n=2$ and $n=4$). Continental subsurface data thus splits between coarse-everywhere maps and exquisite-nowhere research holes (Figure 1); the scalable middle — millions of driller’s logs — is what remains unharmonized. That absence, not a computational or feasibility barrier, is the gap this inventory fills.

A second, less-remarked barrier is one of access position rather than data existence. The comprehensive prior compilations were assembled by or with the agencies that hold the records: Bayless et al. drew on full “State-managed collections” obtained as an inter-agency data request, not a public download. For anyone outside those walls — the modeling and broader science community — the same records are scattered across dozens of independent state portals with no common index, each demanding bespoke reverse-engineering; what is a phone call between agency offices becomes, for the public, a fragmented data-mining problem. We therefore frame this inventory deliberately around the public pathway — every record reachable through a public endpoint with no privileged access — which is both why our per-state point counts can trail an agency’s internal totals and why a reproducible, outside-the-firewall assembly is itself part of the contribution.

The lock is as often legal-institutional as it is technical, and California is the sharpest exemplar. Its driller’s logs were statutorily confidential under California Water Code § 13752 (State of California, 2024) — releasable only to authorized agencies or with the owner’s consent — and a UC Davis analysis notes that until ~ 2017 essentially all California well logs were privately held data (well-attributed but secondary). They became broadly public only

after the Sustainable Groundwater Management Act (2014) and the Open and Transparent Water Data Act (AB 1755, 2016) (State of California, 2016), once owner personally identifying information was redacted and the reports posted, so the bulk dataset is only ~ 8 years old. The same statutory, not physical, barrier recurs across states; the data exists and is increasingly public, but it stayed locked for institutional reasons that newly machine-readable methods (§2.3) are only now able to unlock.

2 Methods

2.1 Harvest — and the heterogeneity it exposes

State sources are ArcGIS REST (most), bulk file, or by-request. A registry-driven harvester paginates each REST layer to FlatGeobuf. We encountered, and the supplement catalogues (Table S1), about ten distinct access modes. The central methodological finding is that the access mode is not predictable from the portal type: an HTML viewer resolved to a structured server-side lithology table for some states (OK, IA) and to scanned PDFs for others (UT, ID); a structured-looking column was 100% empty (KS); and a state booked as carrying one interval per well (IL) in fact exposed the full multi-interval log through a hidden, born-digital text-layer PDF endpoint. Every source must be dereferenced and probed before it is classified.

2.2 Harmonization to the Wellogic schema

Each state’s native lithology is mapped to one canonical interval schema (`source_id | state | well_id | seq | top_ft | bottom_ft | thickness_ft | description | normalized`). Depths are retained in feet (1 ft = 0.3048 m), the unit of the source driller’s logs. Adapters handle separate interval tables, wide inline columns (un-pivoted to long), and multi-file bulk sources. Verbatim driller text is always retained; a rule-based keyword map provides the controlled vocabulary (refinable post hoc). The downstream goal is parity with the per-well

hydraulic attributes that MODFLOW-style groundwater-model construction consumes (horizontal and vertical hydraulic conductivity, aquifer thickness, and static water level); §4.3 reports what raw ingredients exist for that derivation.

2.3 PDF triage and extraction — a full-population, cost-definitive design

For the PDF-only states we do not run a blind full mine, nor rely only on sampling. The classification step—does a log have a text layer?—is free (`pdftotext`) and needs no vision model, so it can be run over the whole state. For each well’s log we stream-and-discard: fetch, run `pdftotext`; if a text layer is present, extract the lithology now via a free state rule-parser (where the layout is a clean table) or a cheap text-only large language model; if the page is a raster scan, the well is recorded to an exact “needs-vision” worklist and deferred. No PDF is retained. This yields, per state, the harvested digital intervals, an exact count of vision-required logs, and therefore a definitive cost (digital cost incurred + vision count $\times$ measured per-log vision cost) rather than an extrapolated estimate. Raster logs are read with a vision-language model (Gemini 2.5, native PDF input, forced JSON schema), an approach grounded in a method base that matured only recently: multimodal models now read text-rich and handwritten documents (Gemini Team, Google, 2024; OpenAI, 2024; Bai et al., 2025; Chen et al., 2024; Liu et al., 2024; Humphries et al., 2024; Crosilla et al., 2025), and the “documents→structured database” paradigm is established in adjacent domains (Dagdelen et al., 2024). Within hydrogeology, machine learning has been used to predict lithofacies from borehole geophysics (Hammond and Allen, 2024), and natural-language processing is only now being introduced to groundwater archives (Christenson and McCoy, 2026); within geoscience more broadly, prior LLM work classifies already-transcribed borehole text (Ghorbanfekr et al., 2025). None recovers structured lithology from scanned driller’s logs at scale. The shift that makes this tractable is recent: vision-language models from 2023 onward collapse OCR, handwriting recognition, layout parsing, and semantic typ-

ing into a single prompt-and-validate step, changing the extraction economics from per-log hand-transcription to metered inference. The closest scanned-record prior art quantifies the pre-VLM barrier and clarifies our methodological departure. Ma et al. (2024) extract well location and depth from 160 historical oil-and-gas records via an OCR-to-text conversion feeding a text LLM (Llama 2): accuracy holds at $\sim 100\%$ on clean digital PDFs but falls to $\sim 70\%$ on image-based scans, precisely where OCR breaks down. Ghorbanfekr et al. (2025)’s domain-adapted model (GEOBERTje) classifies already-transcribed borehole descriptions into lithology classes, outperforming a GPT-4 prompt by 11%, but never touches a scan. We differ on all three axes: a vision-language model reads the scanned form directly (no OCR stage to fail), the target is the harder material-by-depth lithology log rather than point attributes, and it runs over a multi-state corpus rather than hundreds of documents.

3 Results

3.1 Availability census

Of $\sim 13.4\text{M}$ wells across 48 CONUS states, after systematic probing lithology is machine-readable (free, no vision model) for 17 states ($\sim 7.0\text{M}$ wells), available only as scanned PDF logs for 13 states ($\sim 4.0\text{M}$), and without public digital lithology for 18 ($\sim 2.4\text{M}$) (Table 2, Figure 2). A central finding is that the machine-readable category is roughly twice what a surface audit suggests: an initial classification placed only 8 states in category A, but deeper probing revealed structured sources — server-rendered HTML lithology tables (e.g. Tennessee, New Mexico, Vermont), bulk strata files (Texas), and born-digital report PDFs (Ohio, Massachusetts) — for many states first taken to be PDF-only, moving $\sim 3.7\text{M}$ wells from the vision-model column to the free column. Classification is evidence-based and bidirectional: Oklahoma reclassified PDF $\rightarrow$ digital once its viewer was found to render a server-side HTML lithology table (243,801 wells), whereas Kansas reclassified digital $\rightarrow$ PDF once its `GEOLOGICAL_FORMATION` column proved 100% empty. Category C is “no harmonized dig-

Table 2: Availability census of depth-resolved lithology, 48 CONUS states.

Category	States	Wells	Access
A — machine-readable lithology (free)	17	~7.0M	REST / inline / bulk / structured-HTML / text PDF
B — scanned-PDF lithology (vision model)	13	~4.0M	raster scan (this method)
C — no public digital lithology / no DB	18	~2.4M	none / by-request / access- blocked

ital lithology,” not “no data”: a state-by-state regulatory audit (§4.2) found that records in these states are public and almost always merely undigitized (paper/county-health custody) or held behind an unwired portal, rather than legally sealed today. Outright sealing of the report itself is now rare and was largely historical: California’s reports were statutorily confidential until their ~2017 release (§4.1), and the main current restriction we found is Delaware’s withholding of public-supply well locations.

3.2 Harmonized digital inventory

The harmonized v1 release spans 17 states (Table 3): ~24 million depth-resolved lithology intervals across ~3.4 million wells, with Michigan and Texas dominant.

Coverage is far from uniform across states and data elements (Figure 3): lithology dominates the structured states, while the hydraulic and derivation-readiness fields (static water level, depth, yield, surface elevation, completion date) are present in widely varying fractions, and the scanned-only states remain to be extracted.

3.3 Extraction triage: the LLM/VLM split and its cost

An initial per-state calibration (n=100; Table 4, Figure 4) establishes the routing split the rest of the evaluation builds on. A retrievable log is present for 87–100% of wells everywhere; what varies sharply is the vision-required fraction — the share of logs that are raster scans needing a vision model rather than born-digital text — which spans 0–100% and is anti-correlated with e-permitting era: recent e-permitting states (IL, WI, FL) are mostly born-

Table 3: Harmonized digital lithology inventory, v1 release (frozen snapshot, all 17 states).

State	Source	Intervals	Notes
MI	Wellogig	5,473,000	canonical schema (ships H_COND/V_COND/AQ_THK)
TX	SDR	5,289,458	bulk lithology/strata files
MN	CWI C5ST	2,837,410	GeoPackage stratigraphy table
IL	ISGS	2,198,466	text-layer PDF, free
MT	GWIC	1,539,985	100% non-null
CA	OSWCR	1,334,875	4-CSV union; PLSS section-center geometry
WI	WCR	1,261,058	born-digital triage
OK	OWRB	1,099,839	viewer-resolved HTML table
MO	DNR / MSDIS	946,651	wide-column un-pivot
MA	MassDEP	742,424	born-digital report PDFs
VT	ANR	376,867	structured grid
TN	TDEC	309,078	viewer-resolved table
KY	KGS	167,906	structured source
OH	ODNR	162,433	viewer-resolved table
IA	GeoSam	118,503	HTML-table scrape
NM	OSE	52,490	structured API (WAF-limited; partial)
AL	NGWMN	40,710	sparse source (1.7% non-null)
17 states	—	23,951,153	≈3.39M wells total

digital (the free/cheap text route), while legacy paper-archive states (OR, NV, SD, ID) are almost all scans. Extraction is faithful across all states — driller shorthand transcribed verbatim, legibility flagged, and empty arrays returned (rather than fabricated intervals) on blank logs. The definitive large-sample yield and the token-metered cost follow in Section 3.4, and per-well validation against agency transcriptions in Section 3.5.

Definitive cost via full triage. Because classification is free, the full-population triage (§2.3) converts the sampled estimate into a definitive per-state figure while harvesting the digital share for free. Illinois is the first completed case: triaging all 390,443 Illinois wells confirmed 0% vision-required (born-digital text layers), harvesting 2,198,466 lithology intervals from 382,245 wells (≈5.8 intervals/well) at zero vision-model cost. A national projection over the ~4.0M scanned-PDF (category B) wells routed to the vision model — the raster remainder once probing has moved the born-digital share into the free category A — is on the order of 4 billion tokens (≈\$2k at flash-lite); the binding constraints are wall-clock and

Table 4: Initial per-state PDF triage calibration (n=100, Gemini 2.5 flash-lite, sorted by vision share): the born-digital (text-LLM) versus raster (vision-model) routing split. Dollar figures are illustrative; Table 5 gives the definitive large-sample yield and token-metered cost.

State	has-PDF %	text-LLM %	raster-VLM %	intervals/log	\$/1k
SD	100	0	100	10.9	0.37
ID	96	4	96	10.6	0.32
NV	93	5	95	7.2	0.25
OR	100	15	85	4.1	0.19
KS	90	29	71	9.8	0.30
FL	89	67	33	15.0	0.47
WI	98	78	22	3.2	0.20
IL	87	100	0	5.0	0.19

normalization-vocabulary QA, not dollars or storage (stream-and-discard avoids a ~ 700 GB archive).

3.4 Large- n yield and hydraulic-field coverage

We evaluated eleven scanned-PDF states (1,000–1,500 logs each; Table 5, Figures 5 and 6). A single “success rate” conflates three distinct outcomes, so we report them separately. Extraction yield — the fraction of documents the model actually read that yielded a log — is 94.6% pooled (11,806 documents) and 99–100% for most states: once a scan is legible, the model recovers a log essentially always, and it returns an empty array rather than fabricating intervals when none is present. The recovery gaps lie on two other axes. Document availability varies widely: 28% of Mississippi and 63% of Utah wells simply have no scan on file (“no-PDF”). And one state is a genuine extraction outlier — Nebraska (65.5%) — because its modern registrations increasingly omit the lithology log (the temporal heterogeneity of Section 3.5); New York’s apparent shortfall is instead transient server flakiness ($\sim 24\%$ retryable fetch failures), not extraction. For the spatially/temporally heterogeneous states (MS, NE, NV) the reported figure is from the dual-stratified run. We report cost in tokens, the reproducible unit, since per-token prices drift and vary by provider: the VLM (raster) route uses $\approx 1,000$ tokens per log (≈ 600 input — which includes the rendered-page

Table 5: VLM evaluation, eleven scanned-PDF states, reported on three separate axes. Extraction yield = documents-with-a-log / documents-read (Wilson 95% CI); no-PDF = wells with no scan on file; design = age-stratified or dual spatial $\times$ temporal. n_{read} is documents the model read.

State	design	n_{read}	extraction yield % (95% CI)	no-PDF %	int/log
WA	both	1,497	100.0 (99.7–100)	0.1	5.8
SD	age	970	99.9 (99.4–100)	0.1	7.2
ID	age	928	99.9 (99.4–100)	5.0	9.8
NV	both	1,421	99.7 (99.3–99.9)	4.8	9.6
KS	age	872	99.7 (99.0–99.9)	9.1	7.5
UT	both	559	99.6 (98.7–99.9)	62.7	10.1
OR	both	1,491	99.2 (98.6–99.5)	0.3	6.8
MS	both	1,055	100.0 (99.6–100)	28.5	9.0
NY	age	722	94.7 (92.9–96.1)	3.4	9.9
FL	age	861	90.2 (88.1–92.1)	10.3	7.1
NE	both	1,430	65.5 (63.0–67.9)	3.6	8.9
pooled	—	11,806	94.6	—	—

image tokens — plus ≈ 410 output), while the born-digital text route is higher at $\approx 1,800$ tokens/log because the extracted text exceeds a single page image in token count. Across all evaluation logs the spend was ≈ 13 million tokens ($\approx \$3$ derived at the Gemini 2.5 flash-lite list price).

Critically, recording every hydraulic field—returning null when absent—turns the evaluation into a coverage map of what the documents contain (Figure 6). Static water level is present on roughly half of documents (23–61%) and yield on 15–43%, giving a usable hydraulic backbone. But the pump-test fields are sparse—drawdown and pump rate are mostly below 20%—and specific capacity is essentially never written on the forms (0.1–0.5% across all 6,000 documents). This confirms, at scale, that the specific-capacity $\rightarrow$ transmissivity $\rightarrow$ conductivity pathway — the standard route for estimating aquifer hydraulics from well records (Rotzoll and El-Kadi, 2008) — is not broadly recoverable from driller’s logs; derived conductivity is feasible only where yield and drawdown co-occur (here, OR and ID). The zeros are themselves a result: each field was searched in every document, and its absence is now quantified rather than assumed.

3.5 Extraction accuracy: validation against agency transcriptions

Yield (Section 3.4) measures whether a scanned log is readable, not whether the model reads it correctly. California permits a direct accuracy test: 86,653 wells carry both a DWR hand-transcribed FreeForm geologic log and the original scanned report. We sampled 1,499 such wells, passed each scan through the same flash-lite pipeline, and scored its output against the agency transcription as ground truth (Table 6). Agreement is high on every axis that matters: interval count Pearson $r = 0.94$ (means 12.9 vs. 13.1; MAE 0.91 intervals), depth-boundary $F_1 = 0.97$ (± 5 ft; recall 0.98, precision 0.97), and 12-class material accuracy 91.5% (Cohen’s $\kappa = 0.90$) over 516,317 compared feet. On the coarse/fine binary that texture and conductivity models actually consume, agreement is 96.0% ($\kappa = 0.92$). Per-class recall is highest for rock/bedrock (0.96), shale (0.95), sand (0.92) and gravel (0.90), and lowest for the rare silt (0.79) and boulders (0.77). Two residual error sources are reported rather than hidden: (i) $\sim 6\%$ of wells split their log across multiple scanned-page files, and our single-page comparison understates recall for those (a fixable stitching gap that produces the long tail in depth-extent error); and (ii) part of the material “disagreement” is wording granularity — terse human entries (“Soil & Rock”, color-only lines) versus the fuller scan — rather than model error, making 91.5% a conservative floor. Against the agency’s own labels, the model matches coarse/fine texture essentially perfectly and material class to $\kappa \approx 0.90$, validating the recovered layer for hydrogeologic use.

Sampling representativeness. A 1,000-log estimate is only as good as its sample, so we re-evaluated six states ($n=1,500$) under a combined spatial $\times$ temporal stratification (a $\sim 1^\circ$ latitude/longitude grid crossed with completion decade) and tested whether yield is homogeneous across strata. For a proportion, precision depends on the sample size, not the population: at $n=1,000$ the worst-case 95% margin is $\pm 3.1\%$ and the finite-population correction is ≈ 1.000 , so sampling 1,000 of $\sim 10^6$ scans is statistically tight provided the sample is unbiased. Three states (WA, OR, UT) are design-robust — per-stratum yield

Table 6: VLM extraction accuracy versus DWR hand-transcribed geologic logs, California (n=1,499 wells; 516,317 ft of section compared; ground truth = DWR FreeForm transcription).

Metric	Value
Interval count — Pearson r / MAE	0.94 / 0.91
Depth-boundary F_1 (± 5 ft)	0.97
recall / precision	0.98 / 0.97
Logged-depth extent — r	0.95
Material class accuracy (12-class)	91.5%
Cohen’s κ	0.90
Coarse/fine accuracy	96.0%
Cohen’s κ	0.92

spreads ≤ 9 percentage points with all strata mutually inside Wilson 95% intervals. Two are not: Mississippi varies by geography (north-west counties $\sim 50\%$ versus $\sim 86\text{--}89\%$ in the south, tracking per-county scan coverage), and Nebraska and Nevada vary by age — Nebraska’s modern registrations increasingly omit the lithology log (76% recovery in the 1960s falling to 23% in the 2020s). These heterogeneities, invisible to an age-only or spatially-clustered sample, both justify the dual stratification and are themselves findings about how within-state completeness varies. The stratified runs also separate extraction yield from fetch reliability: success as a fraction of documents the model actually read is the design-robust quantity, whereas success as a fraction of attempts also absorbs transient server failures (Washington’s apparent 77% rises to $\sim 99.9\%$ of fetched documents once download flakiness is excluded).

4 Discussion

The contribution is the harmonized framework plus a reproducible, cost-bounded method — not a finished national mine. We need not download and process every log to deliver value: we provide the method, the per-state cost/accuracy/coverage estimates, and the harmonized data we can already release. The same harmonized lithology, paired with the pump-test ingredients that often accompany it, is the input continental models currently approximate

with GLHYMPS-class generalized geology (§1.1), offering a path to replace coarse deep-conductivity units with observation-based stratigraphy where log density allows.

4.1 California: a case study in locked-but-recoverable data

California concentrates both themes of this paper — the data exists but is locked, and vision-language methods newly unlock it — and we treat it as a worked example. The California Department of Water Resources hosts on the order of ~ 1 million scanned Well Completion Reports (our enumeration of the CKAN “WCR PDF Links” resource returns 1,012,784; DWR itself typically states “ $\sim 800,000$ – $1,000,000$ ”, so we treat the figure as approximate), each carrying the driller’s lithology log. Yet only $\sim 104,000$ unique wells’ lithology has been transcribed into DWR’s structured geologic-log tables (across overlapping format categories, not additive: FreeForm $\approx 87,633$, Generalized $\approx 22,260$, QuickPick $\approx 6,800$, USCS $\approx 3,071$), leaving roughly 90% of California’s per-well lithology locked inside un-transcribed scans.

Three compounding reasons kept it locked. (i) Legal confidentiality: the logs were sealed under California Water Code § 13752 (State of California, 2024) and became broadly public only after SGMA (2014) and AB 1755 (2016) (State of California, 2016), so the public bulk dataset is only ~ 8 years old (§1.2). (ii) Manual-transcription cost: even the USGS GAMA program has hand-transcribed only $\sim 19,247$ reports verbatim ($< 3\%$ of the corpus), basin-by-basin (Haugen et al., 2023), the barrier being handwriting and heterogeneous historical forms that classic OCR cannot read. (iii) Aggregate-not-per-well, low ROI: every major use collapsed lithology into smoothed percent-coarse or texture fields rather than per-well logs. The USGS Central Valley Hydrologic Model obtained $> 150,000$ scanned logs from DWR but used only $\sim 8,500$, as binary coarse/fine percent-coarse on a ~ 1 -mile grid (Faunt, 2009; Faunt et al., 2010), following the foundational drillers’-log texture method of Laudon and Belitz (1991); CVHM2 continued this lineage (Marcelli et al., 2022); the attributed-WCR releases carry ~ 260 – 290 k construction attributes but not lithology (Borkovich et al., 2023); and Stanford’s airborne-electromagnetic work used $\sim 11,645$ drillers’-log sediment

profiles only to calibrate airborne EM (Knight et al., 2018; Kang et al., 2025). To our knowledge, no published work performs statewide, per-well lithology extraction from the scanned California WCR corpus, and none applies vision-language models to it — the per-well, full-corpus extraction our triage pipeline (§2.3) is designed to make routine. As a direct test we streamed 1,000 randomly sampled un-transcribed California WCRs through the pipeline: every report fetched as a scanned (raster) PDF, 97.2% yielded at least one lithology interval (mean 7.85 intervals per log), at \$0.29 per thousand logs. At that yield the entire ~908,000-well un-transcribed remainder is recoverable for ≈ 0.9 billion tokens ($\approx \$260$ derived at the current flash-lite list price) — turning a 90%-locked statewide corpus into a structured per-well lithology layer for the token-equivalent of a single field day.

4.2 Why the data is scattered — a governance outcome, not absence

The fragmentation has a specific, documentable cause, and it is not that the data is missing or withheld. Surface-water data is federally unified because one agency both collects and houses it: the USGS National Water Information System, populated through the federal-state Cooperative Water Program, which imposes a single measurement method and schema. Groundwater never acquired such a home. Water wells are regulated by state law—there is no federal well-construction or well-log-reporting standard—because well permitting rides on state primacy over water allocation, itself split between riparian (humid East) and prior-appropriation (arid West) doctrines. The result is fifty schemas at the point of record creation and, within many states, custody divided by agency mission (geological survey vs. water resources vs. environment vs. health). Every federal attempt to unify the records was structurally limited: the National Ground-Water Monitoring Network (SECURE Water Act) is a voluntary aggregator that cannot compel reporting; the Safe Drinking Water Act regulates only public water systems, leaving the tens of millions of domestic wells—where the lithology lives—outside any federal frame; and the National Geothermal Data System (DOE/ARRA,

2009–2014) federated the already-digitized, geothermal- and petroleum-oriented borehole holdings of dozens of participating state geological surveys into a common schema—each survey publishing only its own pre-existing digital subset, of uneven completeness, not a fresh digitization of all wells—proving the integration concept feasible before losing funding and going dark (Allison et al., 2014). Crucially, none of these reached the shallow domestic water-well driller’s logs that carry the lithology: NGDS drew on state geological survey archives rather than the water-agency well-completion systems, ingested existing digital records rather than extracting the scanned backlog, and ran entirely before the largest water-well report corpora were even released to the public (e.g. California’s ~2017 unsealing, §4.1). The barrier is therefore sustainability and scope, not capability—prior efforts showed national harmonization works, but none targeted, reached, or outlived the water-well lithology problem this inventory addresses.

Privacy is real but narrow today, and was historically broader. Modern state public-records law is public-by-default; where records remain restricted it is the owner’s identity (California Water Code §13752 now releases construction and geology while withholding owner name and address) or the well’s exact location (PLSS-grid or landowner-supplied coordinates; Delaware withholds public-supply well locations for source-water security)—not the lithology. The important caveat is temporal: several states’ reports were fully confidential until recently—California’s were statutorily sealed until their ~2017 public release (§4.1)—so confidentiality was a genuine historical brake on bulk extraction, now largely lifted, even though it does not explain the cross-state scatter (which is federalism, above). Domestic well permitting exemptions explain why some logs are never collected at all (Maxwell et al., 2019). Crucially, a state-by-state audit of the apparent “no-data” states found most to be a harvesting miss rather than an absence: of eleven states initially flagged, seven expose completion reports through a named public portal, and the genuine gaps (Georgia, Maryland, Rhode Island, West Virginia) are paper/county-health records never digitized—public records obtainable by request, not confidential. “No digital lithology” therefore means not

yet harmonized or digitized, essentially never absent or legally sealed.

4.3 Known gaps and shortcomings — a living inventory, not a closed product

We catalogue current holes so users can judge fitness and later versions can close them. Lithology coverage is partial: the open v1.0 release harmonizes ~ 3.4 million wells into ~ 24 million depth-resolved intervals (Table 3) — the harvested-to-date subset of the ~ 7 million wells the census finds carry machine-readable lithology. The remainder is recoverable by the same pipeline; the scanned-only states are characterized by validated sampling (yield, accuracy, cost) rather than fully harvested, and a few states publish no public log at all. *Hydraulic ingredients are present but uneven*: static water level and total depth form a broad backbone, but the pump-test fields needed to derive conductivity are scarce — specific capacity is recoverable mainly where yield and drawdown co-occur (e.g. Texas, Arizona, Wisconsin). We separately audited the three derivation-readiness inputs that govern whether transmissivity can be reconstructed downstream: land-surface elevation (to place depth-referenced contacts and water levels on a common datum), measured aquifer-test transmissivity or conductivity, and pump-test duration. The asymmetry is stark. Surveyed land-surface elevation — reported as-is, never substituted with a modeled DEM fill — is bulk-available for eleven states, often complete (Alabama, Iowa, New Mexico at 100%; Ohio 97%, Kentucky 94%, Illinois 86%) but absent from most records elsewhere because elevation, unlike the driller-observed depths and water levels, is not measured at the wellhead. A well completion or construction date fares far better: twenty-two states report it, most at 90–100%, giving the inventory a temporal axis (record vintage) for age-stratified analysis. Pump-test duration — the input that lets specific capacity be promoted to transmissivity via the Theis/Logan relations — is harvestable for Texas ($\sim 156,000$ wells), Kansas, and Nevada. But measured transmissivity exists in public bulk form for essentially a single state (Nevada, $\sim 25,000$ wells) and nowhere else at scale — the strongest evidence that a national conduc-

tivity layer must be derived from the yield/drawdown/duration ingredients rather than read off the records, and feasible only where those ingredients co-occur. Normalization leaves a minority of intervals as “other” (regional vocabulary); geometry is PLSS section-center for some states; and the scanned extraction, although validated against agency transcriptions for $\sim 1,500$ wells (Section 3.5), is not yet run over every state’s full corpus. Some states resist the driller’s-log model entirely: Colorado’s subsurface data is geophysical (gamma/resistivity logs) and aquifer-framework based, not depth-to-material logs. None of these is fatal; each is a documented, improvable hole.

4.4 Transferability beyond the United States

The United States is a demonstration, not the limit of the method. The conditions that make U.S. well lithology hard to assemble — records created under dozens of independent jurisdictions, partially digitized, sometimes behind document portals or legal restriction, and never unified — describe the global state of water-well data, and are typically more severe in regions with the least monitoring capacity. The pipeline depends only on (i) a public list of wells carrying a document link and (ii) a vision-language model; it needs no privileged agency access, no per-form templates, and no country-specific training. The same triage — born-digital text versus raster scan — and the same token-metered extraction apply wherever driller’s or borehole logs are scanned, which includes much of Europe, South Asia, Africa, and Latin America where national well records remain image archives. For data-poor regions, where agency-scale manual transcription is unaffordable, a method whose marginal cost is a few hundred tokens of inference per well lowers the barrier to the subsurface-data foundation that sustainable groundwater management requires. We therefore frame the contribution as a transferable instrument for groundwater-data democratization, validated at scale in one country.

Data availability

A frozen snapshot of the harmonized inventory — the depth-resolved lithology table and the structured hydraulics table — is archived in a public repository (Zenodo) with a citable DOI and is cited in the reference list (Rafei, 2026). Because the upstream public sources mutate and disappear, this frozen version is the authoritative record. The essential method code — the vision-language extraction prompt and output schema, the validation harness, and the harmonization routines — is archived in the same deposit; the live per-state access resolvers are documented in the provenance supplement but are deliberately not distributed as a runnable pipeline, because state endpoints change. An interactive viewer of the inventory is additionally maintained at the project website (<https://swatgenx.com>); the archived DOI, not the website, is the citable source.

Declaration of competing interests

The author develops and maintains SWATGenX (<https://swatgenx.com>), which hosts the optional interactive viewer referenced above; this presents no financial conflict with the findings reported here. The author otherwise declares no competing interests.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used a large language model (Anthropic Claude) to assist with drafting and editing text and code. The author reviewed and edited all content and takes full responsibility for it. Separately, generative vision-language models are employed as a core *research method* for lithology extraction; that use is described in full in the Methods and is distinct from this manuscript-preparation declaration.

CRedit authorship contribution statement

Vahid Rafiei: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Visualization, Writing — original draft, Writing — review & editing.

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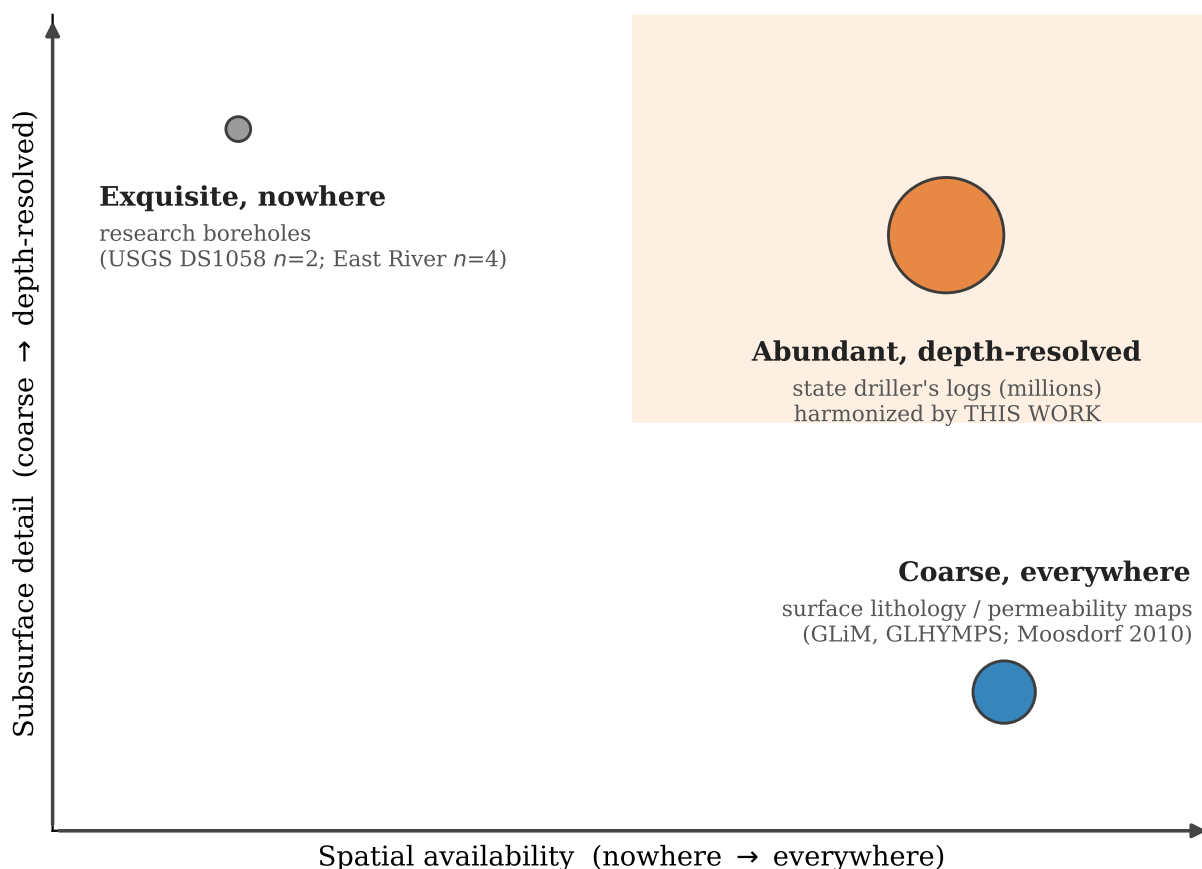


Figure 1: How the CONUS subsurface is represented today, on two axes the existing sources trade off against each other: spatial availability (horizontal) and depth detail (vertical). Surface lithology and permeability maps are coarse but everywhere (lower right); research boreholes are exquisite but almost nowhere (upper left, n of order single digits). Only state driller's logs — millions of them, and depth-resolved — occupy the abundant-and-detailed quadrant (upper right); they are the sole scalable source of three-dimensional subsurface information, and the contribution of this work is to harmonize them. Marker size is schematic, scaled to relative abundance.

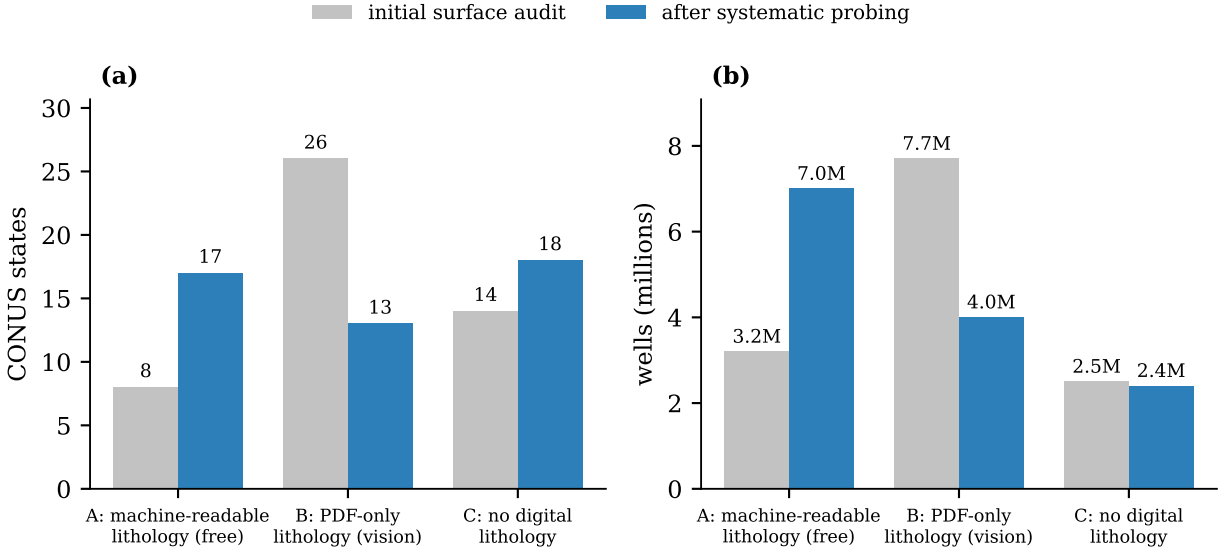


Figure 2: Lithology availability across 48 CONUS states by (a) number of states and (b) number of wells, shown before and after systematic per-state probing (§2.3). The initial surface audit (grey) is what a casual inspection of state portals suggests; systematic probing (blue) is the classification carried forward everywhere else in this paper (Table 2). Probing reclassifies nine states and ~ 3.7 M wells from the vision-model category B into the free, machine-readable category A — the central availability finding. Totals are conserved: 48 states and ~ 13.4 M wells in both columns.

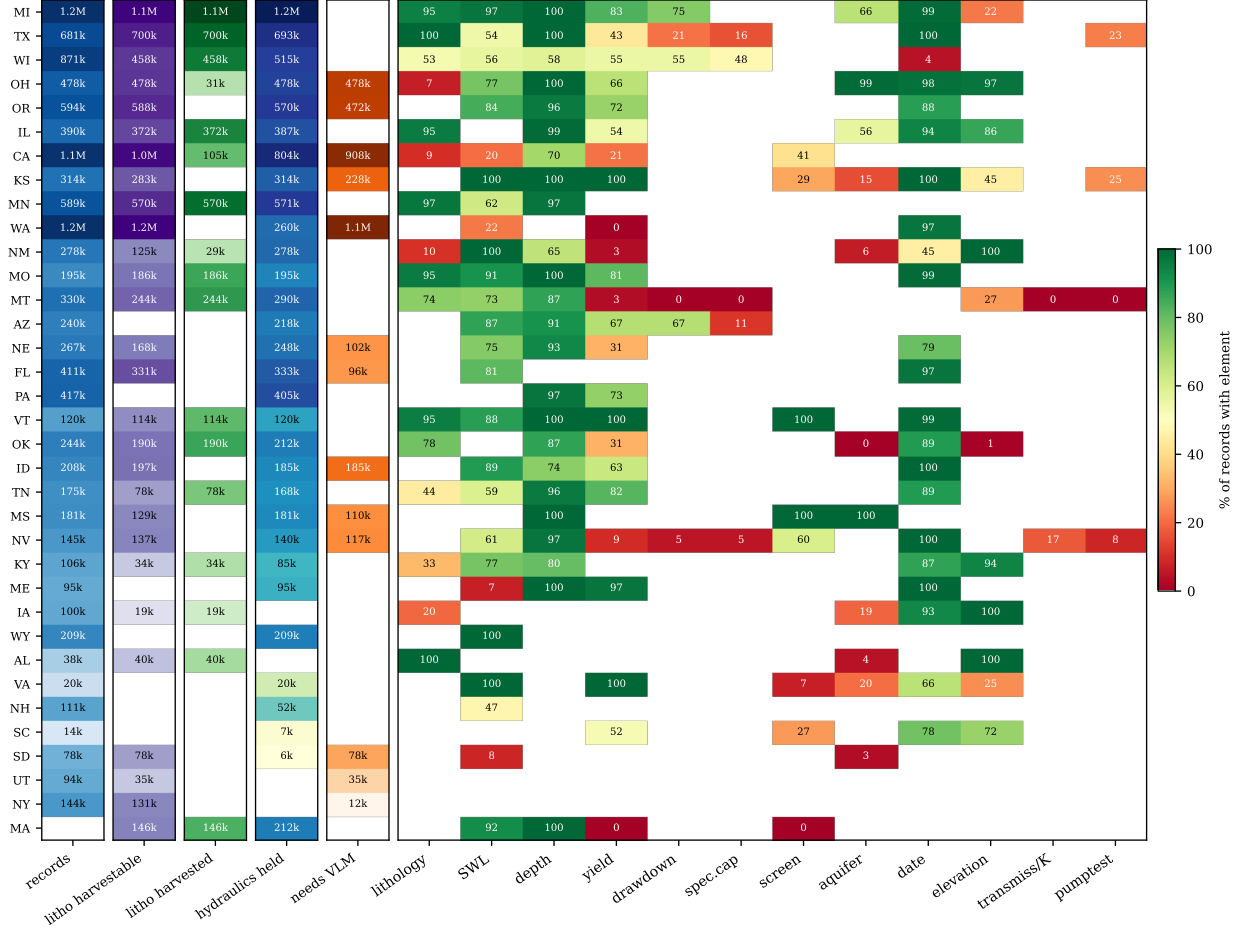


Figure 3: Per-state coverage of the harmonized inventory across the conterminous United States, ranked by estimated count of fully-attributed records. The five count columns are: *records* (wells held); *litho. harvestable* (wells whose lithology is obtainable); *litho. harvested* (wells with lithology now in the inventory); *hydraulics held* (wells with any structured hydraulic value); and *needs VLM* (scanned lithology still to extract). The heatmap reports, for each element, the percentage of a state's records that actually hold that element — harvested lithology plus the structured hydraulics and derivation-readiness fields (static water level, depth, yield, drawdown, specific capacity, screen, aquifer, completion date, surface elevation, measured transmissivity/conductivity, and pump-test duration). Blank cells are not-yet-harvested or not held, not a prediction; Michigan is drawn from the full Wellogic source.

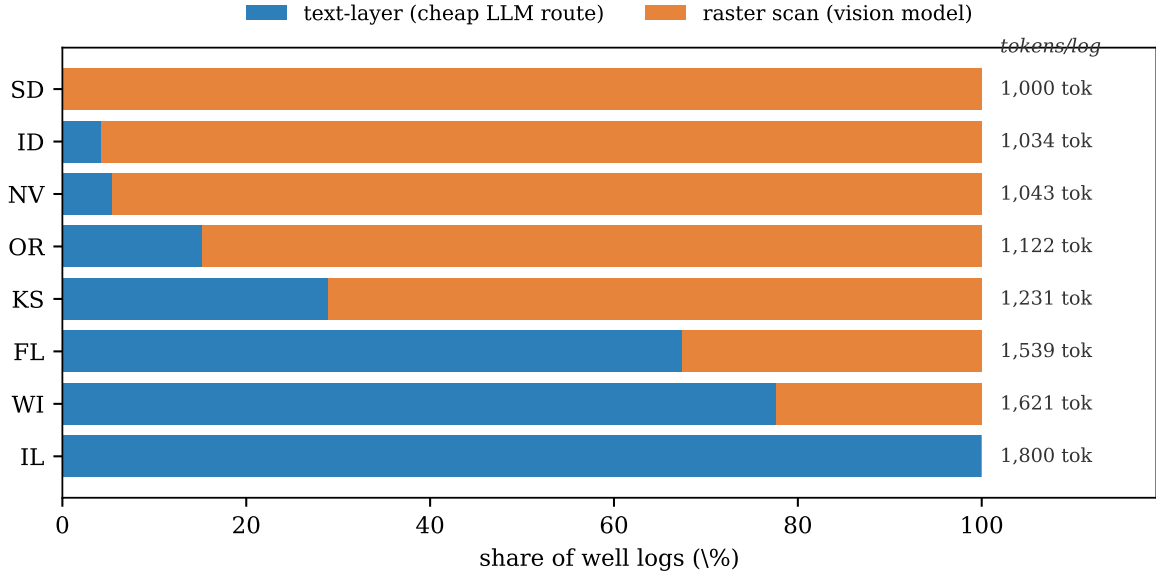


Figure 4: Per-state PDF triage (n=100): the share of well logs that are born-digital (cheap LLM route) versus raster scans (vision model), with the blended token cost per log (text-layer $\approx 1,800$ tokens/log; vision-model raster $\approx 1,000$). The vision-required fraction spans 0–100%, anti-correlated with e-permitting adoption.

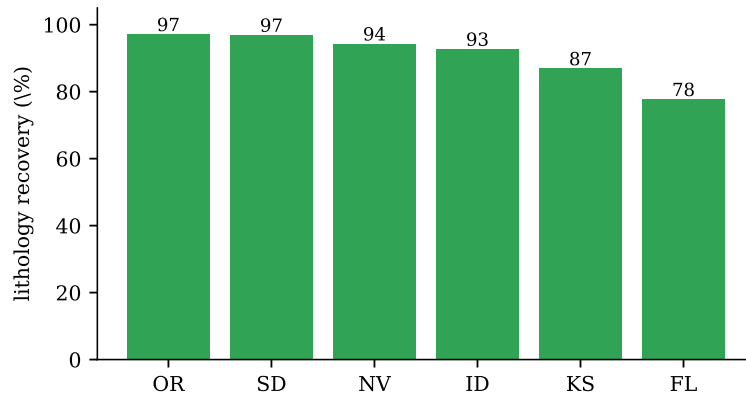


Figure 5: Per-state lithology extraction yield (illustrative; Table 5 gives the authoritative eleven-state values, confidence intervals, and the no-PDF availability axis).

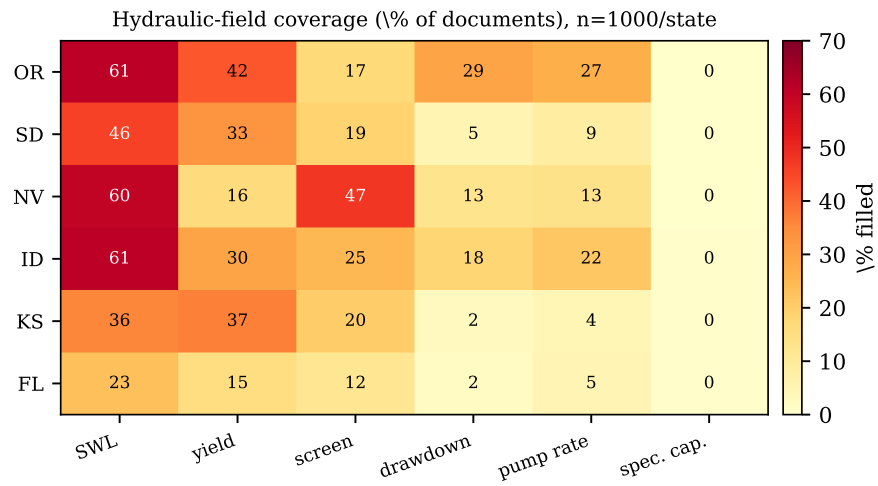


Figure 6: Hydraulic-field coverage: % of documents on which each field is present (n=1,000 per state). Static water level and yield form a backbone; specific capacity is essentially never reported.